Assignment 3

Book Inventory System: Develop a Book class using constructors and exception handling. Store and manage records using ArrayList.

https://github.com/krrish-talesara/Programming-in-Java-Lab-

MainBook

```
import java.util.ArrayList;

public class MainBook {
  public static void main(String[] args) throws InvalidGenreException , InvalidPriceException {
    try {
      Book b1 = new Book("How to get 9 CGPA", "Kavish", "978", "Boring", 500);

      ArrayList<Book> alb = new ArrayList<>(); alb.add(b1);
      Book b2 = new Book("How to get 0 CGPA", "robert", "978", "Tp", 300); alb.add(b2);
      Book b3 = new Book("Harry Potter", "K J Rowling", "978", "Best", 400); alb.add(b3);

      Book b4 = new Book(b3);
      b4.title = "Harry Potter and the Sorcerer's Stone"; b4.price =
      450;
      alb.add(b4);

      Book b5 = new Book("Inferno", "Dan Brown", "978", "Thriller", 350);

      Book b6 = new Book("My experiment with truth", "Mystery", "978", "Biography", -
400);
    ArrayList<Book> alb2 = new ArrayList<>(alb);
    alb2.add(b5);
    alb2.add(b6);
    alb.addAll(alb2);
    System.out.println("Books of genre action");
    alb.forEach(b -> {
      if ("Action".equalsIgnoreCase(b.getGenre())) {
        System.out.println("Title: " + b.getTitle());
        System.out.println("Author: " + b.getAuthor());
        System.out.println("ISBN: " + b.getISBN());
        System.out.println("Genre: " + b.getGenre());
        System.out.println("Price: " + b.getPrice());
        System.out.println();
      }
```

```
});
```

```
        alb.forEach(e -> {int w = ((b1.getPrice()+ b2.getPrice()+ b3.getPrice()+ b5.getPrice()+
b6.getPrice())/5); System.out.println("Average Price: " + w);});
      } catch (InvalidPriceException e) {
        System.out.println(e.getMessage());
      }
    }
}
```

 # Book

```
public class Book { public
    String title; public String
    author; public String ISBN;
    public String genre; public
    int price;

    // Default constructor public
    Book() {
      this.title = "";
      this.author = "";
      this.ISBN = "";
      this.genre = "";
      this.price = 0;
    }

    public Book(String t, String a, String i, String g, int p) throws InvalidPriceException , InvalidGenreException
    {
      this.title = t; if (p
      < 0) {
        throw new InvalidPriceException("Price cannot be negative: " + p);
      }
      if (!g.equalsIgnoreCase("Action") CC !g.equalsIgnoreCase("Biography") CC
!g.equalsIgnoreCase("Best") CC !g.equalsIgnoreCase("Boring") CC
!g.equalsIgnoreCase("Thriller")) {
        throw new InvalidGenreException("Invalid genre: " + g);
      }
      this.author = a; this.ISBN =
      i;
```

```
        this.genre = g; this.price =
        p;
    }

    // Copy constructor public
    Book(Book b) {
        this.title = b.title;
        this.author = b.author;
        this.ISBN = b.ISBN;
        this.genre = b.genre;
        this.price = b.price;
    }

    public String getTitle() {

        return title;
    }
    public String getAuthor() {
        return author;
    }

    public String getISBN() {
        return ISBN;
    }

    public String getGenre() {
        return genre;
    }

    public int getPrice() { return
        price;
    }
}
```

InvalidGenreException

```
public class InvalidGenreException extends Exception { public
    InvalidGenreException(String m) {
        super(m);
    }
}
```

InvalidPriceException

```
public class InvalidPriceException extends Exception { public
  InvalidPriceException(String m) {
    super(m);
  }
}
```

Output)

```
Title: Inferno
Author: Dan Brown
ISBN: 978
Genre: Thriller
Price: 350

Title: My experiment with truth
Author: Mystery
ISBN: 978
Genre: Biography
Price: 400

Average Price: 390
```

```
All Books:
Title: How to get 9 CGPA
Author: Kavish
ISBN: 978
Genre: Boring
Price: 500

Title: How to get 0 CGPA
Author: robert
ISBN: 978
Genre: Best
Price: 300

Title: Harry Potter
Author: K J Rowling
ISBN: 978
Genre: Best
Price: 400
```

Exception Handling:-

1) **Negative Price**

```
Price cannot be negative: -400
(base) PS D:\STUDY\Java Lab\Assignment 3>
```

2) **Invalid Book Genre**

```
Exception in thread "main" InvalidGenreException: Invalid genre: tp
        at Book.<init>(Book.java:25)
        at MainBook.main(MainBook.java:10)
(base) PS D:\STUDY\Java Lab\Assignment 3>
```